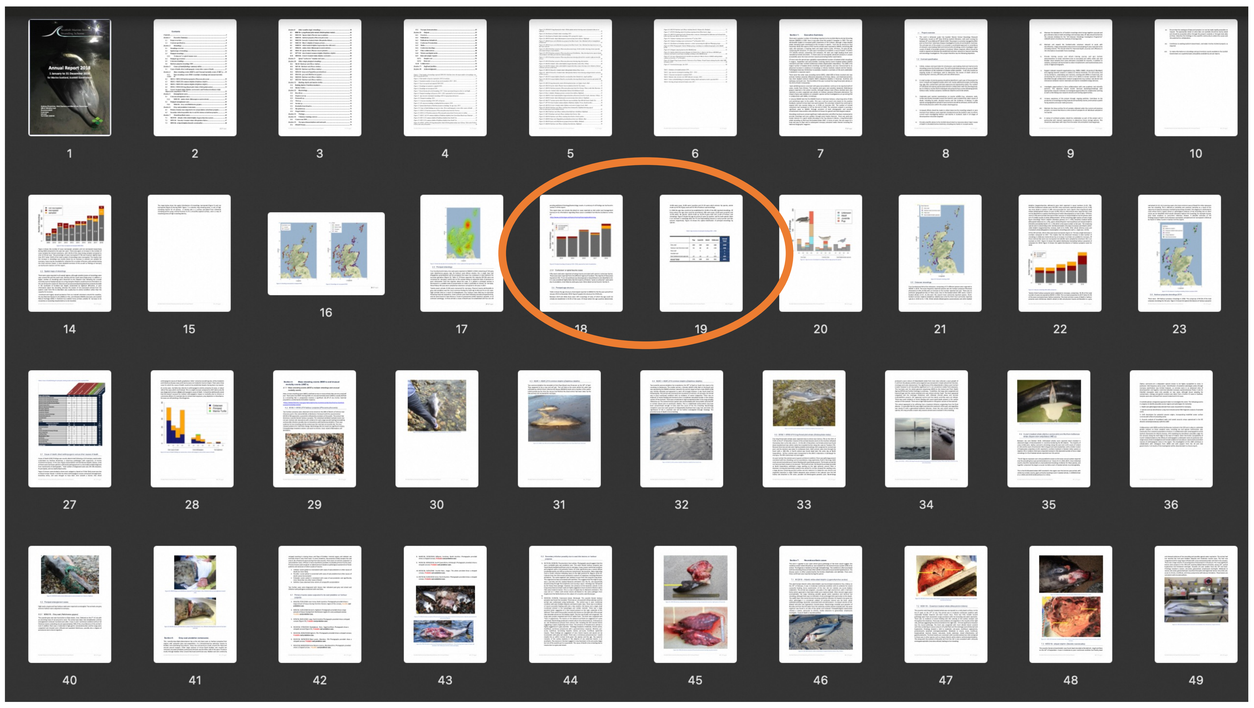

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pending additional histology/bacteriology results. A summary of all findings can be found in section 3 of this report.

This report does not include the detail on cases reported as shot under seal management licences (n=3). Information regarding these cases is available from Marine Scotland or online at:

http://www.scotland.gov.uk/topics/marine/licensing/seallicensing.

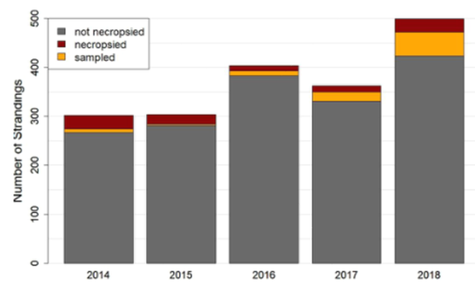

Figure 8: Pinniped strandings (all species) 2014– 2018, separated by level of examination

2.3.1 'Corkscrew' or spiral trauma cases

Fifty-seven seals were reported as having trauma consistent with spiral or corkscrew injuries. These cases were reported from nine different regions of Scotland. The majority of these were Hg seals (n=43): Pv (n=12) and those too decomposed or data deficient to be identified (n=2). It is now considered highly likely that the majority of historic "spiral or corkscrew" cases were due to predation, most likely by adult grey seals. More detail can be found in Section 6.

2.4 Pinniped age structure

Table 2 shows the age structure of pinnipeds reported to SMASS for the five year period from January 2014 to December 2018. Figure 9 graphs the structure for 2018 only.

Between 2014 and 2018, there were 1871 strandings of seals, of which the age could not reliably be established in 50.3% of the cases. Of those where the age could be determined,

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14.8% were pups, 21.8% were juveniles and 13.1% were adult animals. By species, adults made up 16.5% of grey seals and 13.3% of harbour seal strandings.

In 2018 the age class could not be established for 42.8% of the 499 reported strandings. Of those where the age class could be ascertained, 18% were pups; 25.9% were juveniles and 13.3% adults. By species, adults made up 16.5% of grey seals and 11.6% of harbour seal strandings. Figure 9 shows the age structure of cases by quarter, and for both species there is an increase in strandings after the Pv and Hg breeding seasons in early summer and late autumn respectively. Figure 10 shows the spatial distribution of pinniped strandings by species.

Table 2: Age structure of pinniped strandings 2014 – 2018

	Pup	Juvenile	Adult	Unknown	Grand Total
Grey seal	200	309	182	408	1099
Harbour seal (Common seal)	52	82	45	159	338
Harp seal		1			1
Hooded seal				1	1
Seal (indeterminate species)	27	16	17	372	432
Grand Total	276	408	244	940	1871

Question: What is the total number of pinniped strandings reported between 2014 and 2018?
Ground Truth: 1871

GPT-4o ✗
Response: The total number of pinniped strandings reported between 2014 and 2018 can be found in the bar graph in the 9th image. The numbers for each year are as follows:
- 2014: Approximately 150
- 2015: Approximately 150
- 2016: Approximately 200
- 2017: Approximately 200
- 2018: Approximately 500

Adding these together gives a total of approximately 1200 pinniped strandings reported between 2014 and 2018.
Final Answer: 1200

Gemini-1.5-Pro ✓
Response: Between 2014 and 2018, a total of 1871 pinniped strandings were reported. This is shown in Table 2 on page 19.
Final Answer: 1871

Qwen2-VL ✗
Response: The total number of pinniped strandings reported between 2014 and 2018 is 690.
Final Answer: 690

LLaVA-OneVision-Chat ✗
Response: The total number of pinniped strandings reported between 2014 and 2018 is 1,386.
Final Answer: 1386

Qwen-VL-Max ✗
Response: To determine the total number of pinniped strandings reported between 2014 and 2018, we need to sum the number of strandings for each year within this range.

From the bar chart in the image, we can see the following data for pinniped strandings:
- 2014: Approximately 300 strandings
- 2015: Approximately 350 strandings
- 2016: Approximately 400 strandings
- 2017: Approximately 450 strandings
- 2018: Approximately 500 strandings

Adding these numbers together gives us the total number of pinniped strandings reported between 2014 and 2018:
300 (2014) + 350 (2015) + 400 (2016) + 450 (2017) + 500 (2018) = 1900 strandings
Therefore, the total number of pinniped strandings reported between 2014 and 2018 is 1900.
Final Answer: 1900

LLaVA-Next-Interleave ✗
Response: The total number of pinniped strandings reported between 2014 and 2018 is 1,372.
Final Answer: 1372

Error Analysis: Wrong Evidence Sources.
Details: Relevant evidence can be found in both the table in page 19 and the bar chart in page 18, but the total num can be extracted directly from the table in page 19. Only Gemini-1.5-Pro get the right answer. GPT-4o and Qwen-VL-Max struggle to estimate values from the bar chart. Another three open-source models response with wrong short answer.